SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
DEPARTMENT OF ARTIFICIAL INTELLIGENCE
B. Tech-3rd Year (5th Sem)
Natural Language Processing (AI357)
Mid Semester Exam

Date: 24/09/2026 Time: 11:15 AM To 12:45 PM Marks: 6*5=30

1. You are building a morphological analyzer for nouns in English language using a finite state transducer. ‘s’ suffix is concatenated to regular nouns to create plural forms. But there is an exception to it when the following occurs:
   A noun ending in s, x, z, ch, or sh forms its plural by adding es.
   Design an FST that can produce all singular and plural forms of nouns as stated above. Use +SG for singular features, +PL for plural features, and +N for noun features. Show all states and transition table as well. [CO1] [6]

2. Consider the following corpus:
   low lower low lower lowest
   low lowest low lowest
   low new
   new new newer newer
   new

   Apply the wordpiece algorithm for 2 merges. What will be tokenization of the word *newest* ? [CO2] [2+2+2=6]

3. Consider the following sentences in the toy dataset:
   cat dog animal
   cat cat pet
   dog pet animal
   cat dog pet

   Represent each word in terms of a TF-IDF score. Use raw counts for TF and unsmoothed IDF while computing TF-IDF. Which two sentences are closest to each other? Show all computations. Don’t add any start and end sentence markers. [Hint: Use TF-IDF vectors to represent each sentence.] [CO3] [4+2=6
   ]

4. Design a Naive Baiyes classifier to detect whether a mail is spam or not on the following data. *free*, *win*, and *offer* columns represents binary features whether this word appears in the email or not.

Email	**free**	**win**	**offer**	**Class**
E1	1	1	1	Spam
E2	1	1	0	Spam
E3	1	0	1	Spam
E4	0	1	1	Spam

E5	0	0	1	NotSpam
E6	0	1	0	NotSpam
E7	1	0	0	NotSpam

What will be your prediction for new email E8 with features (0, 0, 0) where the features are in the same order as the above dataset? Show the complete computations. [CO4] [4+2=6]

5. Consider the following 5 sentences. Don't add any start and end sentence markers.
I like NLP
I like AI
You like NLP
You love AI
We love NLP

Use Kneser-Ney smoothing to find out P_{KN}(AI | love) and P_{KN}(NLP | like). Use d = 0.5 in your computations. [CO1+CO2] [3+3=6]